Retarded GF | (Read Ch. 2 of Rickayzen's book!)

$$G_R(t,t') = -i \langle [A(t), B(t')]_\varepsilon \rangle \, \theta(t-t') =$$
$$= -i \, \mathrm{Tr}\{ \rho_0 [A(t), B(t')]_\varepsilon \} \, \theta(t-t')$$

full density matrix of the system

$$[A(t), B(t')]_\varepsilon \equiv A(t) B(t') + \varepsilon B(t') A(t)$$

$$\varepsilon = \begin{cases} -1, & \text{if } A, B \text{ are Bose operators} \\ +1, & \text{if } A, B \text{ are Fermi operators} \end{cases}$$

$$\Rightarrow [A, B]_\varepsilon = \begin{cases} [A, B] = AB - BA & \text{for } \varepsilon = -1 \\ \{A, B\} = AB + BA & \text{for } \varepsilon = +1 \end{cases}$$

Advanced GF

$$G_A(t,t') = i \langle [A(t), B(t')]_\varepsilon \rangle \, \theta(t'-t)$$

Causal GF

$$G_c(t,t') = -i \langle \hat{T} A(t) B(t') \rangle =$$
$$= -i \, \theta(t-t') \langle A(t) B(t') \rangle + i \varepsilon \, \theta(t'-t) \langle B(t') A(t) \rangle$$

where $\langle A(t) B(t') \rangle \equiv \mathrm{Tr}\{ \rho_0 A(t) B(t') \}$.

Thermal (temperature) GF

$$G(\tau, \tau') = - \langle \hat{T}_\tau A(-i\tau) B(-i\tau') \rangle$$
$$= - \theta(\tau - \tau') \langle A(-i\tau) B(-i\tau') \rangle + \varepsilon \, \theta(\tau' - \tau) \langle B(-i\tau') A(-i\tau) \rangle$$

Here $A(t) = e^{iHt} A e^{-iHt} \Rightarrow A(-i\tau) = e^{iH(-i\tau)} A e^{-iH(-i\tau)} =$
$$= e^{H\tau} A e^{-H\tau}.$$

Spectral representation

Consider retarded GF of operators $A(t), B(t')$:

$$G_R(t) = -i\,\theta(t) \langle [A(t), B(0)] \rangle = -i\theta(t)\, \mathrm{Tr}\{\rho_0 (A(t)B(0) + \varepsilon B(0)A(t))\}.$$

Time correlation function

Then $\langle A(t) B(0) \rangle \equiv \mathrm{Tr}\{\rho_0 A(t) B(0)\} = \left(\rho_0 = e^{-\beta H}/Z\right) =$

$$= Z^{-1}\, \mathrm{Tr}\{e^{-\beta H} e^{iHt} A e^{-iHt} B\} =$$

$$= Z^{-1} \sum_n \langle n | e^{-\beta H} e^{iHt} A e^{-iHt} B | n \rangle$$

where $H|n\rangle = E_n |n\rangle$ is a complete set of eigenstates $|n\rangle$.

$$\langle A(t) B(0) \rangle = Z^{-1} \sum_{mn} \langle n | B e^{-\beta H} | m \rangle \langle m | e^{iHt} A e^{-iHt} | n \rangle$$

$$= Z^{-1} \sum_{mn} e^{-\beta E_m} \langle n | B | m \rangle \langle m | A | n \rangle e^{i(E_m - E_n)t}$$

Spectral density

let $J_1(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} \langle A(t) B(0) \rangle =$

$$= 2\pi Z^{-1} \sum_{mn} e^{-\beta E_m} \langle n | B | m \rangle \langle m | A | n \rangle\, \delta(E_m - E_n + \omega).$$

Now, $\langle B(0) A(t) \rangle = Z^{-1}\, \mathrm{Tr}\{e^{-\beta H} B e^{iHt} A e^{-iHt}\}$

$$= Z^{-1} \sum_{nm} \langle n | e^{-\beta H} B | m \rangle \langle m | e^{iHt} A e^{-iHt} | n \rangle =$$

$$= Z^{-1} \sum_{nm} e^{-\beta E_n} \langle n | B | m \rangle \langle m | A | n \rangle e^{i(E_m - E_n)t}$$

so that

$$J_2(\omega) = \int_{-\infty}^{\infty} dt\, e^{i\omega t} \langle B(0) A(t) \rangle = 2\pi Z^{-1} \sum_{nm} e^{-\beta E_n} \langle n | B | m \rangle \cdot$$

$$\cdot \langle m | A | n \rangle\, \delta(E_m - E_n + \omega)$$

Due to delta-function, $E_n = E_m + \omega \Rightarrow e^{-\beta E_n} = e^{-\beta E_m} e^{-\beta\omega}$:

$$\Rightarrow \quad J_2(\omega) = e^{-\beta\omega} J_1(\omega).$$

Thus for the retarded GF

$$G_R(t) = -i\,\theta(t)\left\{\langle A(t)B(0)\rangle + \varepsilon\langle B(0)A(t)\rangle\right\}$$

$$= -i\,\theta(t)\int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\, e^{-i\omega' t}\left(J_1(\omega') + \varepsilon J_2(\omega')\right) =$$

$$= -i\,\theta(t)\int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\, e^{-i\omega' t}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')$$

F.T. $$G_R(\omega) = \int_{-\infty}^{\infty} dt\, G_R(t)\, e^{i\omega t} =$$

$$= -i\int_0^{\infty} dt\, e^{i\omega t}\int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\, e^{-i\omega' t}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')$$

$$= -i\int\frac{d\omega'}{2\pi}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')\underbrace{\int_0^{\infty} dt\, e^{i(\omega-\omega'+i\delta)t}}_{\frac{i}{\omega-\omega'+i\delta}}$$

$$G_R(\omega) = \int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\,\frac{\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')}{\omega-\omega'+i\delta}$$

Note that if $B = \psi$ and $A = \psi^+$ (which is frequent case)

$$J_1(\omega) = 2\pi Z^{-1} \sum_{m,n} e^{-\beta E_m} \underbrace{\langle n|\psi|m\rangle\langle m|\psi^+|n\rangle}_{|\langle n|\psi|m\rangle|^2 = \text{real}} \delta(E_m - E_n + \omega)$$

$J_1(\omega)$ is real function, $J_1(\omega) \geq 0$.

$\Rightarrow J_2(\omega) = e^{-\beta\omega} J_1(\omega) \geq 0$ too.

$$\Rightarrow \operatorname{Im} G_R(\omega) = \operatorname{Im} \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{(1+\varepsilon e^{-\beta\omega'}) J_1(\omega')}{\omega - \omega' + i\delta} =$$

$$= \frac{-\pi}{2\pi} (1+\varepsilon e^{-\beta\omega}) J_1(\omega)$$

Similarly, $\operatorname{Im} G_A(\omega) = \frac{1}{2}(1+\varepsilon e^{-\beta\omega}) J_1(\omega)$

For the advanced GF

$$G_A(t) = i\,\theta(-t)\left\{ \langle A(t)\,B(0)\rangle + \varepsilon \langle B(0)\,A(t)\rangle \right\}$$

$$= i\,\theta(-t)\int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\, e^{-i\omega' t}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')$$

F.T. $$G_A(\omega) = \int_{-\infty}^{\infty} dt\; G_A(t)\, e^{i\omega t} = i\int_{-\infty}^{0} dt\; e^{i\omega t}\,\cdot$$

$$\cdot \int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\, e^{-i\omega' t}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega') =$$

$$= i\int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega') \underbrace{\int_{-\infty}^{0} dt\; e^{i(\omega-\omega'-i\delta)t}}_{\frac{-i}{\omega-\omega'-i\delta}}$$

$$= \int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\,\frac{\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')}{\omega-\omega'-i\delta} = \left(G_R(\omega)\right)^*$$

$$\Rightarrow \mathrm{Re}\,G_A = \mathrm{Re}\,G_R\,;\quad \mathrm{Im}\,G_A = -\mathrm{Im}\,G_R$$

$$\Rightarrow G_R(\omega) - G_A(\omega) = \int_{-\infty}^{\infty}\frac{d\omega'}{2\pi}\left(1+\varepsilon e^{-\beta\omega'}\right) J_1(\omega')\,\cdot$$

$$\cdot \underbrace{\left(\frac{1}{\omega-\omega'+i\delta} - \frac{1}{\omega-\omega'-i\delta}\right)}_{=-2\pi i\,\delta(\omega-\omega')} = (-i)\left(1+\varepsilon e^{-\beta\omega}\right) J_1(\omega).$$

$$\Rightarrow \mathrm{Re}\,G_R + i\,\mathrm{Im}\,G_R - \left(\mathrm{Re}\,G_R - i\,\mathrm{Im}\,G_R\right) = 2i\,\mathrm{Im}\,G_R(\omega) =$$

$$= -i\left(1+\varepsilon e^{-\beta\omega}\right) J_1(\omega).$$

Temperature GF

$$G(\tau) = -\theta(\tau) \langle A(-i\tau)\, B(0)\rangle + \varepsilon\, \theta(-\tau) \langle B(0)\, A(-i\tau)\rangle$$

Choose $0 < \tau \leq \beta \Rightarrow -\beta < (\tau-\beta) \leq 0$, so that

$$G(\tau-\beta) = \varepsilon\, \theta(\beta-\tau) \langle B(0)\, A(-i(\tau-\beta))\rangle =$$

$$= \varepsilon\, Z^{-1} \mathrm{Tr}\left\{ e^{-\beta H} B\, e^{H(\tau-\beta)} A\, e^{-H(\tau-\beta)} \right\}$$

$$= \varepsilon\, Z^{-1} \mathrm{Tr}\left\{ A e^{-H\tau + H\beta} e^{-\beta H} B\, e^{H\tau} e^{-\beta H} \right\} =$$

$$= \varepsilon\, Z^{-1} \mathrm{Tr}\left\{ e^{-\beta H} A\, \underbrace{e^{-H\tau} B\, e^{H\tau}}_{B(i\tau)} \right\} =$$

$$= \varepsilon \langle A\, B(i\tau)\rangle = \varepsilon\, Z^{-1} \mathrm{Tr}\left\{ e^{-\beta H} e^{H\tau} A\, e^{-H\tau} B \right\}$$

because $e^{-\beta H} e^{H\tau} = e^{H\tau} e^{-\beta H}$

$$= \varepsilon \langle A(-i\tau)\, B\rangle = \varepsilon\left(-G(\tau)\right)$$

<u>Fermions</u> $(\varepsilon = +1)$

$$G(\tau-\beta) = -G(\tau)$$

$G_{\text{Fermi}}(\tau)$ is periodic in τ with period 2β.

Expand $\quad G(\tau) = \frac{1}{\beta} \sum_n e^{i\omega_n \tau} G(\omega_n)$

$$G(\tau-\beta) = \frac{1}{\beta} \sum_n e^{i\omega_n(\tau-\beta)} G(\omega_n) = -$$

$$= -\frac{1}{\beta} \sum_n e^{i\omega_n \tau} G(\omega_n)$$

$$\Rightarrow e^{-i\omega_n \beta} = -1 \Rightarrow i\omega_n \beta = (2n+1)\pi, \quad \omega_n = \frac{(2n+1)\pi}{\beta}$$

<u>fermions</u>

<u>Bosons</u> $(\varepsilon = -1) \Rightarrow G(\tau - \beta) = G(\tau)$

now we need $e^{-i\omega_n \beta} = +1 \Rightarrow i\omega_n \beta = 2\pi n$

$$\omega_n = \frac{2n\pi}{\beta} \quad \underline{\text{bosons}}$$

$\Rightarrow G_{\text{Boson}}(\tau)$ is periodic in τ with period β.

For both cases we can write $e^{-i\omega_n \beta} = -\varepsilon$.

Now we consider

$$G(\omega_n) = \int_0^\beta d\tau \, e^{+i\omega_n \tau} \, G(\tau) =$$

$$= -\int_0^\beta d\tau \, e^{i\omega_n \tau} \, Z^{-1} \, \mathrm{Tr}\{e^{-\beta H} e^{\tau H} A e^{-\tau H} B\} =$$

$$= -\int d\tau \, e^{i\omega_n \tau} \, Z^{-1} \sum_{nm} \langle n | B e^{-\beta H} | m \rangle \langle m | e^{\tau H} A e^{-\tau H} | n \rangle$$

$$= -\int_0^\beta d\tau \, e^{i\omega_n \tau} \, Z^{-1} \sum_{m,n} e^{-\beta E_m} \langle n | B | m \rangle \langle m | A | n \rangle e^{\tau (E_m - E_n)}$$

$$= -Z^{-1} \sum_{m,n} e^{-\beta E_m} \langle n | B | m \rangle \langle m | A | n \rangle \frac{e^{(E_m - E_n + i\omega_n)\beta} - 1}{E_m - E_n + i\omega_n}$$

but $e^{i\omega_n \beta} = -\varepsilon$

$$= +Z^{-1} \sum_{m,n} \langle n | B | m \rangle \langle m | A | n \rangle \frac{e^{-\beta E_m} + \varepsilon e^{-\beta E_n}}{E_m - E_n + i\omega_n}$$

Now observe that

$$\int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{J_1(\omega')}{i\omega_n - \omega'} = \frac{2\pi Z^{-1}}{2\pi} \sum_{mn} e^{-\beta E_m} \langle n|B|m\rangle\langle m|A|n\rangle \cdot \frac{1}{i\omega_n + E_m - E_n}$$

since $\displaystyle\int_{-\infty}^{\infty} d\omega' \frac{\delta(E_m - E_n + \omega')}{i\omega_n - \omega'} = \frac{1}{i\omega_n - (E_n - E_m)}$

$$\int \frac{d\omega'}{2\pi} \frac{J_2(\omega')}{i\omega_n - \omega'} = \frac{2\pi Z^{-1}}{2\pi} \sum_{mn} e^{-\beta E_n} \langle n|B|m\rangle\langle m|A|n\rangle \cdot \frac{1}{i\omega_n - (E_n - E_m)}$$

$$\Rightarrow G(\omega_n) = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{J_1(\omega') + \varepsilon J_2(\omega')}{i\omega_n - \omega'} =$$

$$= \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{(1 + \varepsilon e^{-\beta\omega'}) J_1(\omega')}{i\omega_n - \omega'}$$

which should be compared with $G_R(\omega)$:

$$G_R(\omega) = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \frac{(1 + \varepsilon e^{-\beta\omega'}) J_1(\omega')}{\omega + i\delta - \omega'}$$

$\Rightarrow G(\omega_n) = G_R(\omega = i\omega_n)$ for $\underline{\underline{\omega_n > 0}}$ (since $\delta > 0$)